SHIVARAMAN T

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Profile Summary

B.Tech ECE (AI & Cybernetics) student with expertise in **embedded systems, biomedical signal analysis, and real-time data acquisition**. Skilled in **Python, SciPy, FFT, filtering, waveform analysis, and feature extraction**. Hands-on experience developing sensor-driven systems, performing data preprocessing, validating hardware performance, and analyzing physiological time-series datasets.

Education

Vellore Institute of Technology — Sept 2023 – May 2027
B.Tech in Electronics and Communication Engineering (AI & Cybernetics)
CGPA: 8.66 / 10

Technical Skills

Programming: Python, Java, Embedded C
Signal Processing: Butterworth Filtering, FFT, Noise Reduction, Peak Detection, Waveform Analysis, Feature Extraction
Data Analysis: NumPy, Pandas, SciPy, Matplotlib, Statistical Analysis, Time-Series Processing
Embedded Systems: ESP32, Arduino, ADC/DAC Interfaces, PWM Control, Sensor Integration
Core Concepts: Signals & Systems, Control Systems, Digital Electronics, Biomedical Instrumentation
Tools: MATLAB, Simulink, Proteus, LTspice, Git

Experience

Aqualis Inspection Services, Dubai — Engineering Intern — May 2025 – Jun 2025

- Performed verification and validation (V&V) of embedded monitoring systems using real-time sensor acquisition.
- Investigated signal-quality issues and improved operational reliability through debugging and testing.
- Conducted subsystem validation, fault analysis, and technical documentation for compliance workflows.

Projects

Smart Aeroponics System — Real-Time Monitoring & Control — GitHub

- Engineered ESP32-based automation platform for monitoring 4 environmental parameters in real time.
- Improved sensor accuracy by 20% through calibration, filtering, and optimized ADC sampling.
- Designed anomaly-detection workflow using FFT-based signal analysis techniques.
- Reduced nutrient dosing latency using closed-loop feedback and automated control logic.
- Enhanced operational stability through structured hardware validation and testing.

Automatic Aerator System (AquaDox) — GitHub

- Developed intelligent dissolved oxygen monitoring platform using embedded sensor acquisition.
- Increased measurement consistency using filtering and signal-conditioning algorithms.
- Designed automated aerator-control mechanism with threshold-based decision logic.
- Optimized sampling intervals to reduce unnecessary aerator runtime and power consumption.
- Performed long-duration validation to verify stable real-time operation.

ECG Signal Processing and Heart Rate Analysis — GitHub

- Analyzed ECG waveform datasets using Python, NumPy, Pandas, SciPy, and Matplotlib.
- Applied Butterworth filtering and denoising methods to suppress baseline drift and motion artifacts.
- Developed R-peak detection pipeline for accurate heart-rate estimation from physiological signals.
- Extracted waveform features and performed FFT-based frequency-domain analysis.
- Improved signal clarity through preprocessing and noise-reduction techniques.

Smart Patrol Bot — Autonomous Embedded System — GitHub

- Built ESP32-CAM based surveillance robot with PIR-enabled motion detection functionality.
- Developed real-time navigation and obstacle-detection workflow using sensor fusion techniques.
- Improved movement stability through PWM-based motor control optimization.
- Integrated alert-notification system for continuous remote monitoring and security response.

Publications

Smart Helmet Systems: IoT Enhanced Safety and Integrated Display for Two Wheelers
6th International Conference on Recent Advancements in Mechanical Engineering (ICRAME 2025), NIT Silchar

Achievements

- Global Rank 5135 — TCS CodeVita Season 13
- Top 100 — Zealestra X AWS Machine Learning Challenge
- Selected Participant — Bharatiya Antariksh Hackathon 2024 (ISRO)

Certifications

Arduino Programming & Embedded Systems (GUVI - HCL) • MATLAB (Signal Processing) • Embedded Systems (Raspberry Pi & Arduino) • **AWS Certified Cloud Practitioner**